

Control of *Listeria monocytogenes* biofilm on industrial surfaces by cell-free extracts of *Lactobacillus plantarum*

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Abstract

In this study, the anti-microbial and anti-biofilm potential of the cell-free extract (CFE) of *Lactobacillus plantarum* was assessed against *Listeria monocytogenes* 32 isolated from the food sample. CFE of *L. plantarum* showed high anti-microbial activity (15.5 ± 1.4 mm diameter zone) against *L. monocytogenes* 32. The anti-biofilm forming activity with the sub-minimum inhibitory concentration (MIC) value of CFE (15.625 mg/L) for 24 hr co-incubation period was obtained with the $90 \pm 7.9\%$ inhibition. Co-incubation assay was also carried out on different surfaces such as stainless steel, polypropylene, and polyvinyl chloride. Viable cells in the formed biofilm were counted and the greatest reduction ($5.22 \log \text{CFU/cm}^2$) was achieved on polyvinyl chloride surface. Both biofilm formation and inhibition were also visualized by Scanning Electron Microscopy. It was observed that the biofilm of *L. monocytogenes* 32 was thread-like, binding one cell to another and the surface by the extracellular polymeric substance.

Practical applications

The dairy, meat, poultry, and seafood industries are the main food sectors where the biofilm formation of food-borne pathogens is observed. *Listeria monocytogenes* is one of the most noteworthy among them. High costs, procurement difficulties, or resistance problems limit the practical use of several approaches focused on the inhibition of *L. monocytogenes* biofilms. To prevent biofilm formation and/or to destroy cells embedded in biofilm via lactic acid bacterial strains and/or their metabolites has been remarkable in recent years. In the light of data obtained in this study, the cell-free extract of *L. plantarum* with its anti-microbial and anti-biofilm activity may be used to design as a bio-control strategy with its safe and environmentally friendly properties for prevention of biofilm formation of *L. monocytogenes*.

1 | INTRODUCTION

Listeria monocytogenes has been a significant research field for biofilms in the food industry (Abdalla, 2013). Biofilms produced by *L. monocytogenes* with a simpler design than many other microorganisms are considered to be very important in terms of their resistance and industrial problems (Abdalla, 2013; Chae & Schraft, 2000; Kalmokoff et al., 2001). Many various food

processing inert surface types such as stainless steel, glass, plastic, rubber, and polypropylene are ideal environments where *L. monocytogenes* can rapidly and firmly adhere and grow (Chae et al., 2006; Ciccio et al., 2012; Hood & Zottola, 1997; Moltz & Martin, 2005). Growth temperature and surface types are the main factors that significantly change the biofilm production ability of *L. monocytogenes* strains. To give an example, *L. monocytogenes* has a rapid adherence capacity to stainless steel surfaces than

others with its ability to reach an irreversible stage in a few hours (Oh & Marshall, 1996). Besides, recent finding stating the survival ability of *L. monocytogenes* cells showed that the surface colonization ability of *L. monocytogenes* when stored at 4°C or 10°C for at least 5 days which is the low temperature used for processing and storing a significant foodstuff amount in the food industry, increased the propensity for cross-contamination to foodstuffs (Ciccio et al., 2012; Smith et al., 2004). The ability of *L. monocytogenes* to form biofilms on different food processing surfaces at least partially formed the mechanisms causing *L. monocytogenes* survival under harsh physical and chemical stress conditions (Chae et al., 2006; Ciccio et al., 2012).

Different physical, chemical, and biological methods have been used to eliminate biofilms in the food industry (Ciccio et al., 2012). An effective biofilm control method could be provided through chemical treatment, such as sodium hypochlorite and hydrogen peroxide (Srey et al., 2013). Examples of chemical or biological agent alternatives suggested removing biofilms have been accepted as environment-friendly control strategies like enzymes and phages (Srey et al., 2013), natural compounds such as flavonoids extracted from plants (Koo et al., 2003), and bacterial products such as bacteriocins (Okuda et al., 2013). In particular, several studies (Nostro et al., 2016; Sandasi et al., 2010; Santos et al., 2018) have reported the activity of plant extracts in inhibiting the biofilm formation of *L. monocytogenes*. However, no unique strategy providing absolute biofilm control efficiency has been presented in the literature yet (Ciccio et al., 2012). Final tendencies toward the discovery of resistant pathogens such as *L. monocytogenes* in the food chain strengthened the need for searching different alternatives for disinfection. Therefore, there is a great interest in developing novel strategies for surface or equipment-associated pathogen persistence control through natural products, especially in the food industry (Winkelströter et al., 2013). Based on the competitive exclusion principle, using probiotics or their metabolites for inhibition, the colonization and proliferation of pathogens to the hard surfaces have drawn attention and accepted as a promising approach for the control of biofilm formation (Falagas & Makris, 2009; Gómez et al., 2016; Hibbing et al., 2010). Also, it was shown in several studies that the bactericidal effect of biocides on bacterial biofilms was improved by bacteriocin-producing lactic acid bacteria (Gómez et al., 2012, 2016; Lobos et al., 2009). Researchers stated that these strains have a big potential in controlling *L. monocytogenes* biofilms from the point of food processing environments. Furthermore, biofilms formed by lactic acid bacteria present in foods may present promising counteracting means for pathogenic biofilm formation (Gomez et al., 2016; Pérez Ibarreche et al., 2014; Winkelströter et al., 2013).

Application of bacteriocins and/or bacteriocin-producing strains to prevent biofilm formation and/or to demolish biofilm buried cells has been regarded as a new research field. This study aimed to determine the inhibitory effect of cell-free extract (CFE) of *L. plantarum* on the biofilm formed by *L. monocytogenes* strains isolated from food samples and, thus, to develop an alternative bio-control strategy for biofilm control, considering the recent developments in the bio-control field.

2 | MATERIALS AND METHODS

2.1 | Bacterial strains and their cultivation

The investigation was conducted using 20 strains of *Listeria monocytogenes* which were isolated from different poultry (turkey and chicken meat) samples and identified by molecular methods based on 16S rRNA gene sequence analysis with previous studies. In addition, potential pathogenicity features (hemolysis activity, CAMP test, and PI-PLC activity), as well as genotypic virulence properties of these strains (PCR-based serotyping and pathogenic genes-hly, plcA, prfA, iap, inlA, and actA), were also determined (Unpublished data). All strains of *L. monocytogenes* were cultured in Brain Heart Infusion (BHI, Merck) medium at 37°C for 24 hr. *L. plantarum* isolated from herbs (fermented in brine) materials using in herby cheese-making process and identified by MALDI-TOF analysis was used for the determination of its anti-biofilm activity against *L. monocytogenes* strains. The strain was cultured in Man, Rogosa, and Sharpe (MRS, Merck) broth medium at 37°C for 24 hr. All bacterial strains were maintained at -80°C in the appropriate cultivation broth containing 20% (v/v) glycerol.

2.2 | Assessment of anti-microbial activity

CFE of *L. plantarum* was used for the assessment of its anti-microbial activity. To obtain CFE, the strain was grown in MRS broth at 37°C for 24 h and the culture broth was centrifuged at 20,000 ×g for 10 min at 4°C. The culture supernatant was then filtered through the 0.22-μm membrane filter to remove the bacteria cells and was used as the source of CFE. In addition, its neutralized form with 8M NaOH (pH 7.0) and proteinase K treatment (1 mg/ml) was also tested (Saad et al., 2015). Anti-microbial activity of CFE and its treated forms against different *L. monocytogenes* strains were determined by agar well diffusion assay (Savadogo et al., 2004). *L. monocytogenes* strains, grown in BHI agar plates for 24 hr at 37°C, were adjusted to McFarland 0.5, then pour plated on agar surfaces via inoculated soft agar. CFE and its treated samples were transferred as 50 μl into the 6-mm-diameter wells opened on agar surfaces and incubated 24 hr at 37°C. During the experiment, the MRS medium was used as a negative control. Following the incubation, the inhibition zones around the wells were checked and the diameters were measured as mm. *L. monocytogenes* strains in which the maximum anti-microbial activity observed was chosen for further studies. Thirty milliliters of filter-sterilized CFE samples were lyophilized (Christ, Alpha 1-2 LD Plus, UK) under 0.0010 mbar at -80°C for 48 hr. In order to determine the CFE concentrations used in biofilm experiments, MIC (Minimum Inhibitory Concentration) values of the lyophilized CFE (100–1.56 mg/L) against *L. monocytogenes* strains were determined by using the microtiter plate assay method. The lowest concentration of CFE, which prevents visible growth of *L. monocytogenes*, was determined as the MIC value (CLSI, 2012).

2.3 | Biofilm formation of *L. monocytogenes* strains

Biofilm formation was measured by crystal violet assay using polystyrene-based 96 well microplates (LP Italiana) (Stepanovic et al., 2000; Vestby et al., 2009). *L. monocytogenes* overnight cultures, grown in BHI broth at 37°C for 24 hr, were adjusted to McFarland 0.5 and added to the microplate wells. Following the 24 hr incubation at 37°C, the medium was removed and the wells were washed with phosphate-buffered saline (PBS, Merck) three times and fixed by 95% methanol (Merck) for 20 min. Wells were stained using crystal violet solution (0.1%, Merck) and the dye was recovered with acetone:ethanol (30:70, Merck) solution. Biofilm biomass was quantified by spectrophotometry at 595_{nm}. Un-inoculated BHI medium was used as a negative control. The biofilm production capabilities of the *L. monocytogenes* strains were evaluated by calculating the “cut off” values. The optical density cut-off (ODC) was defined as the mean OD value of the negative control. The final OD value of a tested strain was expressed as the average OD value of the strain was reduced by ODC value. Based on the OD, strains were classified as nonbiofilm producers (OD ≤ ODC), weak biofilm producers (ODC < OD ≤ 2 × ODC), moderate biofilm producers (2 × ODC < OD ≤ 4 × ODC), or strong biofilm producer (4 × ODC < OD) (Borges et al., 2012).

2.4 | Anti-biofilm forming activity of *Lactobacillus plantarum*

For determination of the anti-biofilm forming activity of *L. plantarum*, its CFE at a sub-minimum inhibitory concentration (sub-MIC/MIC₅₀) and each of the *L. monocytogenes* strains adjusted to McFarland 0.5 were co-incubated in polystyrene-based 96 well microplate wells for 24 hr at 37°C. Following the incubation, wells were aspirated and planktonic cells were removed, and biofilm formation was analyzed according to the crystal violet assay, as described before. In all assays; an un-inoculated BHI medium was used as a negative control and an inoculated BHI medium without CFE was used as a positive control. Inhibition of biofilm formation was calculated as % inhibition with the formula of [(C-B)-(T-B)]/[[(C-B)] × 100 (C: well including the pathogen; B: well including the medium; T: well including the *L. monocytogenes*; and CFE at MIC₅₀ value together) (Karaca et al., 2015). *L. monocytogenes* strain which of its biofilm formation was mostly inhibited was chosen for further studies.

2.5 | Biofilm formation on different surfaces and determination of the cell viability in the formed biofilm

Different coupons made from 316L stainless steel (SS), polyvinyl chloride (PVC), and polycarbonate (PC) sized in 2.5 × 0.8 × 0.1 cm dimensions with a total surface area of 4 cm² were used as

substrates for biofilm formation. The coupons were treated with isopropanol and boiled for 15 min in a detergent solution and then rinsed five times in distilled water to remove any residual detergent. Then, they were sterilized by autoclave before using. Each coupon was placed into separate wells of a sterile polystyrene-based 12 well plate and biofilm formation and co-incubation assays were carried out on these different types of surfaces, as described before.

In addition to these experiments, the cell viability within the biofilms formed on these different surfaces was also assessed. Following each of the treatments, coupons were washed twice with PBS and then air-dried. For homogenization, all the coupons were vortexed speedily with the glass beads (r:3 mm) for 1 min, and then each of the homogenized solutions was serially diluted and spread plated on BHI agar plates. After 48 hr incubation at 37°C, viable cells of *L. monocytogenes* in the biofilms was calculated as log CFU/cm² for each treatment (Giaouris et al., 2005).

2.6 | Scanning Electron Microscopy (SEM) analysis

After biofilm formation and co-incubation assays, each of the coupons was removed from the well and washed by sterile PBS three times to remove the medium and nonadhered cells. Following the fixation of the coupons at room temperature in 4% formaldehyde for 3 hr, the washing step was repeated and the coupons were dehydrated in serial dilutions of 30, 50, 60, 70, and 90% ethanol for 5 min each, followed by three rinses for 10 min in 100% ethanol (Kiran et al., 2015). Finally, each of the coupons was air-dried and immediately sputter coated with gold (WD = 50 mm, 90 s, 30 mA), then examined using Scanning Electron Microscope (Quanta 400F Field Emission, SEM) at Central Laboratory of Middle East Technical University (Ankara, TURKEY).

2.7 | Statistical analysis

All assays were carried out with three independent experiments performed in triplicate. SPSS version 19.0 (IBM, New York, USA) was used to analyze the data. All results were given as mean ± standard deviation and *p*-value of less than .05 was used to identify a significant difference.

3 | RESULTS AND DISCUSSION

L. monocytogenes can persist and adapted in different food environments for long periods with its potential biofilm formation properties. Biofilm structure is the key factor for the contamination of food products which leads to critical problems in terms of public health and economic impact (Pilchova et al., 2014). However, control of elucidating *L. monocytogenes* and its contamination routes in food-processing environments can be a major challenge, and biofilm

formation is one of the most important aspects of this problem which always remains difficult. There are various chemical, physical, and green technology approaches that aimed to prevent or control the biofilm formed (Oloketuyi & Khan, 2017). Biocides (sodium hypochlorite and benzalkonium chloride), chlorine-based disinfectants are commonly used in the food industry for their bactericidal and anti-biofilm activities (Rodríguez-Melcón et al., 2019; Waghmare & Annapure, 2015). In recent years, probiotic bacteria or their metabolites are used as anti-bacterial agents against some food-borne bacteria (Rao & Sreenivasa, 2016). The objective of this study was to evaluate not only the anti-microbial effect of CFE derived from *L. plantarum* but also its anti-biofilm effect against *L. monocytogenes*, as an important disease-borne bacterium.

3.1 | Assessment of anti-microbial activity of *L. plantarum*

At the beginning of this study, we investigated the anti-microbial activity of CFE derived from *L. plantarum* against 20 different *L. monocytogenes* strains isolated poultry (turkey and chicken meat) samples. Our results revealed that CFE of *L. plantarum* showed anti-microbial

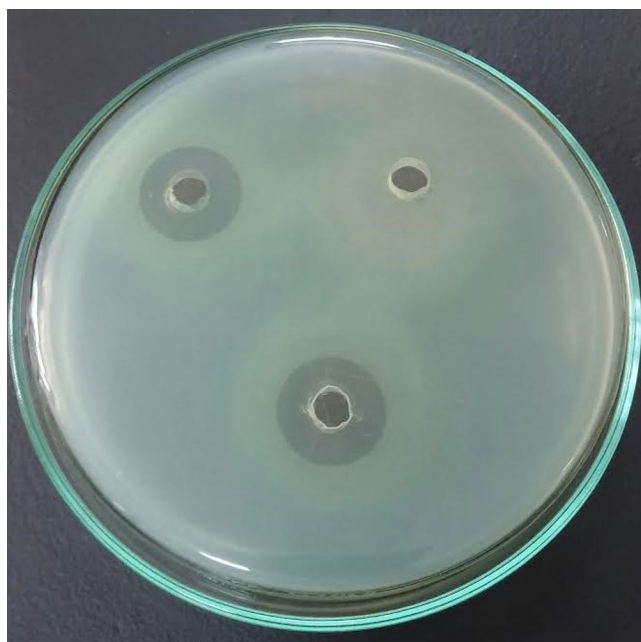


FIGURE 1 Anti-microbial activity of *Lactobacillus plantarum*-derived CFE against *L. monocytogenes* 32 on an agar plate (a): CFE; (b): neutralized CFE; (c): proteinase K treated CFE

activity against all *L. monocytogenes* strains tested. However, five of them that we observed the highest activities with the inhibition zones over 10 mm were selected to be included in the biofilm studies (data not shown). The highest anti-microbial activity (15.5 ± 1.4 mm diameter zone) was observed against *L. monocytogenes* 32. To understand the nature of the activity, a neutralized and proteinase-K-treated form of CFE was also tested. As it was shown in Figure 1, neutralization did not affect the activity, while proteinase-K treatment caused the loss of its activity. There was no significant difference ($p > .05$) in the size of inhibition zones between the CFE and its neutralized form (Table 1). These results indicated that the anti-microbial effect may be due to bacteriocin production as described in many other studies (Pérez-Ibarreche et al., 2017). MIC values of the CFE were found as 31.25 mg/L for all five investigated strains of *L. monocytogenes* and sub-minimum inhibitory concentration (sub-MIC) of CFE (15.625 mg/L) was used for further analysis. Our results represented in this part of the study were found in agreement with results obtained by other studies used different strains of lactic acid bacteria as bio-control agents against *L. monocytogenes* (Arena et al., 2016; Kiran & Osmanagaoglu, 2014).

3.2 | Biofilm formation of *L. monocytogenes* strains and anti-biofilm forming activity of *L. plantarum*

The biofilm formation abilities of five *L. monocytogenes* strains which of the anti-microbial activity were observed were evaluated by calculating the cut-off values. Based on the OD, strains of *L. monocytogenes* 5 and 11 were found as weak biofilm producer with 0.012 and 0.075 OD value while the strains of *L. monocytogenes* 21, 32, and 33 were found strong biofilm producer with 0.48, 0.57, and 0.54 OD values, respectively (Figure 2). The differences between biofilm formation abilities of strains might be explained by the relationship between the ability to form biofilms interaction and phylogeny which was reported by Djordjevic et al. (2002). Similarly, Doijad et al. (2015) was also reported that the correlation between serotype and biofilm formation remained inconclusive and it is dependent on individual strains. As a result of biofilm formation abilities, *L. monocytogenes* 32 was selected among the three strain for further analysis since the highest anti-microbial activity (15.5 ± 1.4 -mm-diameter zone) was detected for this strain as well as being determined as a strong biofilm producer (0.57 value). To prove it, *L. monocytogenes* 32 was co-incubated with the sub-MIC value of *L. plantarum*-derived CFE, biofilm inhibition was observed in polystyrene-based plate surface

Anti-microbial activity

Inhibition zone of CFE (mm)	15.50 ± 1.40^a
Inhibition zone of neutralized CFE (mm)	13.75 ± 1.25^a
Inhibition zone of proteinase K-treated CFE (mm)	0 ± 0.00^b

TABLE 1 Anti-microbial activity of *L. plantarum*-derived CFE against *L. monocytogenes* 32

^aNote Means with different letters in the same column are significantly different ($p < .05$).

FIGURE 2 Biofilm formation of *L. monocytogenes* strains using crystal violet assay

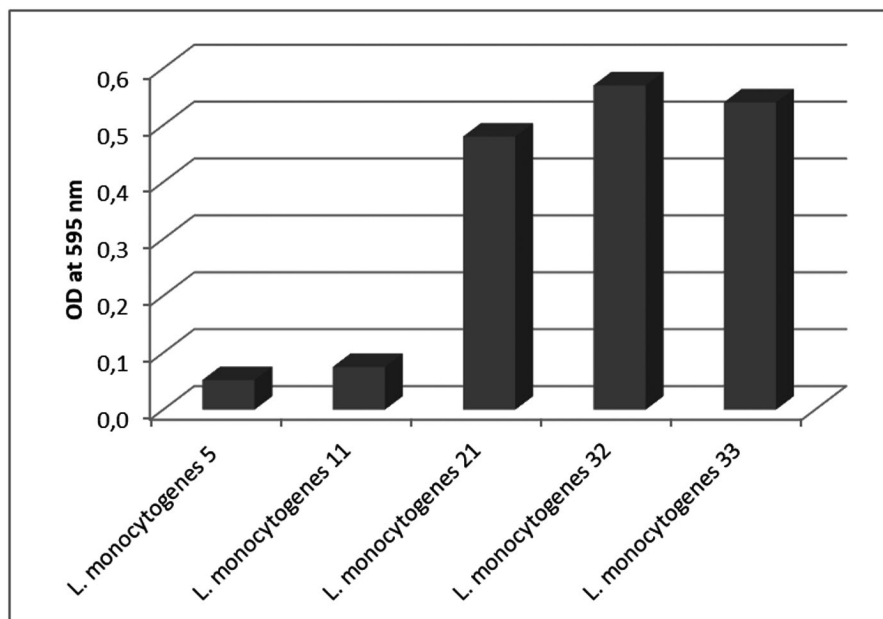


TABLE 2 Anti-biofilm forming activity of *L. plantarum* against *L. monocytogenes* 32 on different surfaces

Surfaces	Inhibition on biofilm formation (%)
316L Stainless steel surface (SS)	78 ± 5.0 ^a
Polyvinyl chloride surface (PVC)	81 ± 4.7 ^a
Polycarbonate surface (PC)	77 ± 3.6 ^a

^aNote Means with same letters in the same column are not significantly different ($p > .05$).

with the inhibition of $90 \pm 7.9\%$. Therefore, *L. monocytogenes* 32 was selected for subsequent anti-biofilm assays.

3.3 | Determination and visualization of the biofilm and anti-biofilm formation on different surfaces

Different coupons made from SS, PVC, and PC were used as food-contact surfaces for biofilm formation of *L. monocytogenes* 32. Following the co-incubation with *L. plantarum*-derived CFE, biofilm formation was expressed as % inhibition. As a result shown in Table 2, no significant differences were observed between the SS, PVC, and PC surfaces. Similarly, Santos et al. (2018) showed that all concentrations tested a fraction of heather extracts led to statistically significant inhibition of biofilm formation of *L. monocytogenes*, with inhibition percentages as high as 91%. Overall, *L. plantarum*-derived CFE with its sub-MIC values was found to be effective on all surfaces which are all resistant to acid and other sanitizer treatments (van der Veen & Abee, 2011) that makes routine disinfection

in the food industry more difficult. However, bioactive compounds are important candidates for the prevention of biofilms development on surfaces that come into contact with foods. Similar to our results; nisaplin adsorbed food-contact surfaces as stainless steel, polyethylene terephthalate, and rubber reduced the adhesion ability of food-isolated *L. monocytogenes* strains (Guerra et al., 2005). Another investigation about the efficacy of nisin in reducing the biofilm formation of *L. monocytogenes* on different surfaces was conducted by Arevalos-Sánchez et al. (2012), and the results indicated the efficacy of nisin on glass and stainless steel surfaces. In addition, Hibma (1997) was reported that the bred phage successfully inhibited the biofilm formation of *L. monocytogenes* on stainless steel.

The types of polymeric surfaces are responsible for the morphology, structure, and functional integrity of the biofilm. Although it is well accepted that *L. monocytogenes* form biofilms on different food-processing surfaces, there has been very little data from the point of microscopical evidence to support this. In this study, scanning electron micrographs demonstrating the presence of individual cells and biofilm formation of *L. monocytogenes* 32 on various surfaces have also been reported and the results confirmed the biofilm formation of *L. monocytogenes* 32 and its inhibition by CFE. The microscopic observations performed for *L. monocytogenes* 32 biofilms showed a dense three-dimensional structure in all surfaces (Figures 3a–5a), similar to the reports of other authors (Borucki et al., 2003; Dubravka et al., 2009). Using the scanning electron microscope, we established that *L. monocytogenes* 32 formed strong biofilm and that individual bacterial cells adsorbed to all three surfaces. The ability of *L. monocytogenes* to form moderate-to-strong biofilms has been also determined by other studies on stainless 311 steel (Rieu et al., 2008), glass (Bridier et al., 2010), or plastic, including polystyrene (Chavant et al., 2002; Guilbaud et al., 2015). Similar to Chavant et al. (2002) and Borucki et al. (2003), *L. monocytogenes*

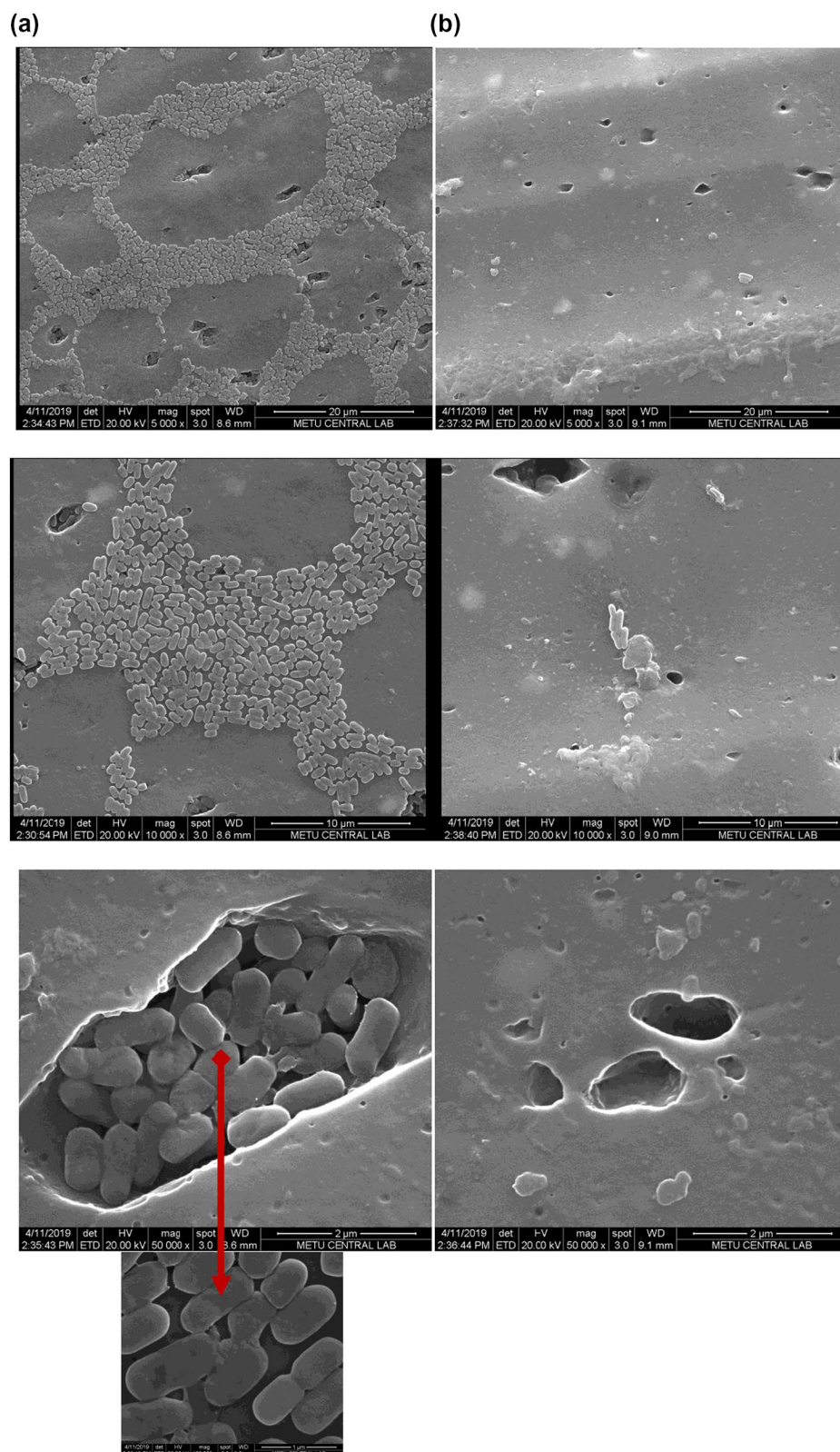


FIGURE 3 Scanning electron micrographs illustrating the biofilm formation of *L. monocytogenes* 32 on 316L stainless steel surface (column A: untreated; column B: treated with the sub-MIC value of CFE)

cells were shown in a cellular network with channels and pores between cell aggregates on all surfaces. Our results have been also correlated with the appearance of the biofilm of *L. monocytogenes*

32 as a “honeycomb” pattern, described by Marsh et al. (2003). It was also observed that the biofilm of *L. monocytogenes* 32 was thread-like, binding one cell to another and the surface by the

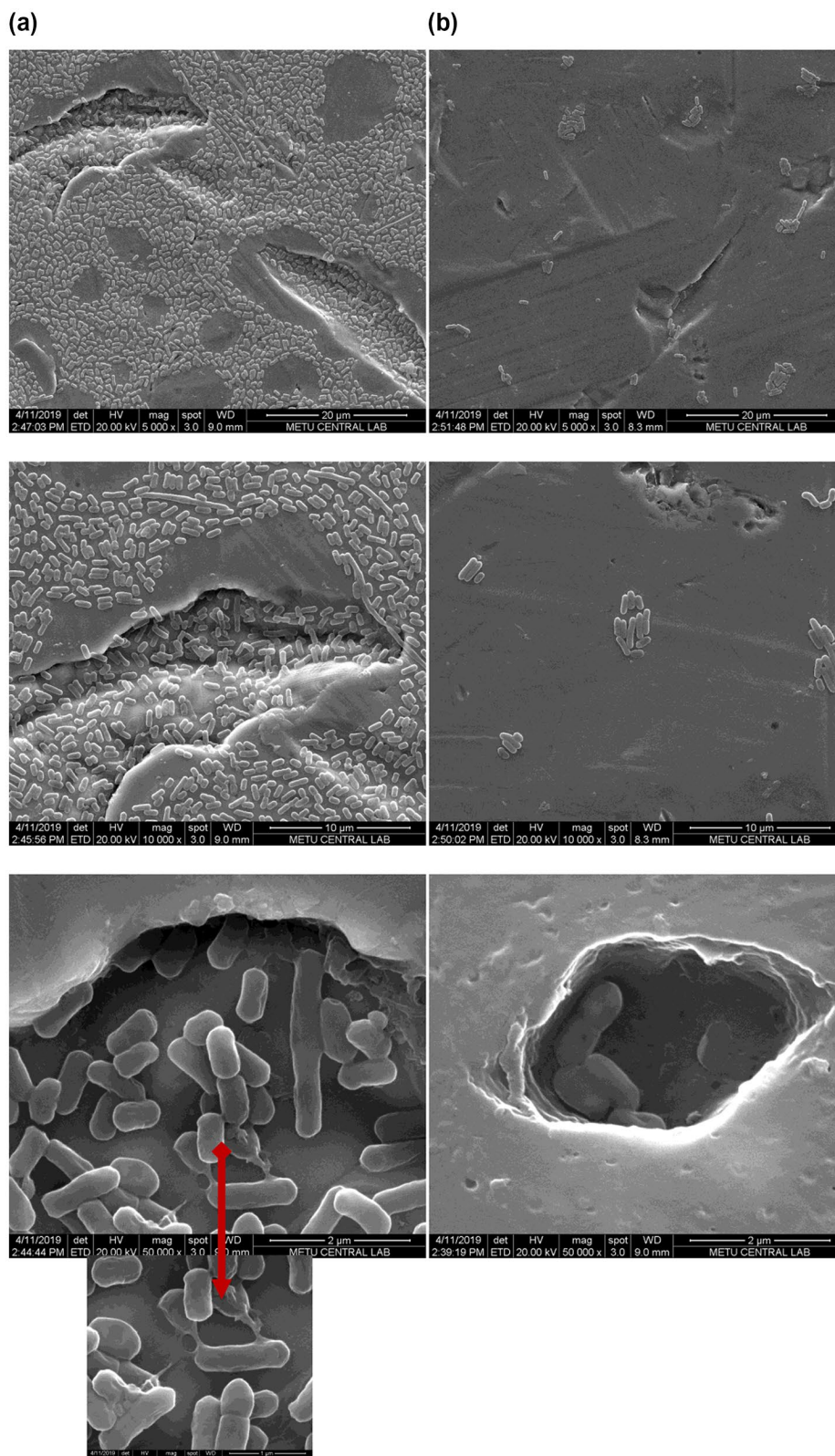


FIGURE 4 Scanning electron micrographs illustrating the biofilm formation of *L. monocytogenes* 32 on polyvinyl chloride surface (column A: untreated; column B: treated with the sub-MIC value of CFE)

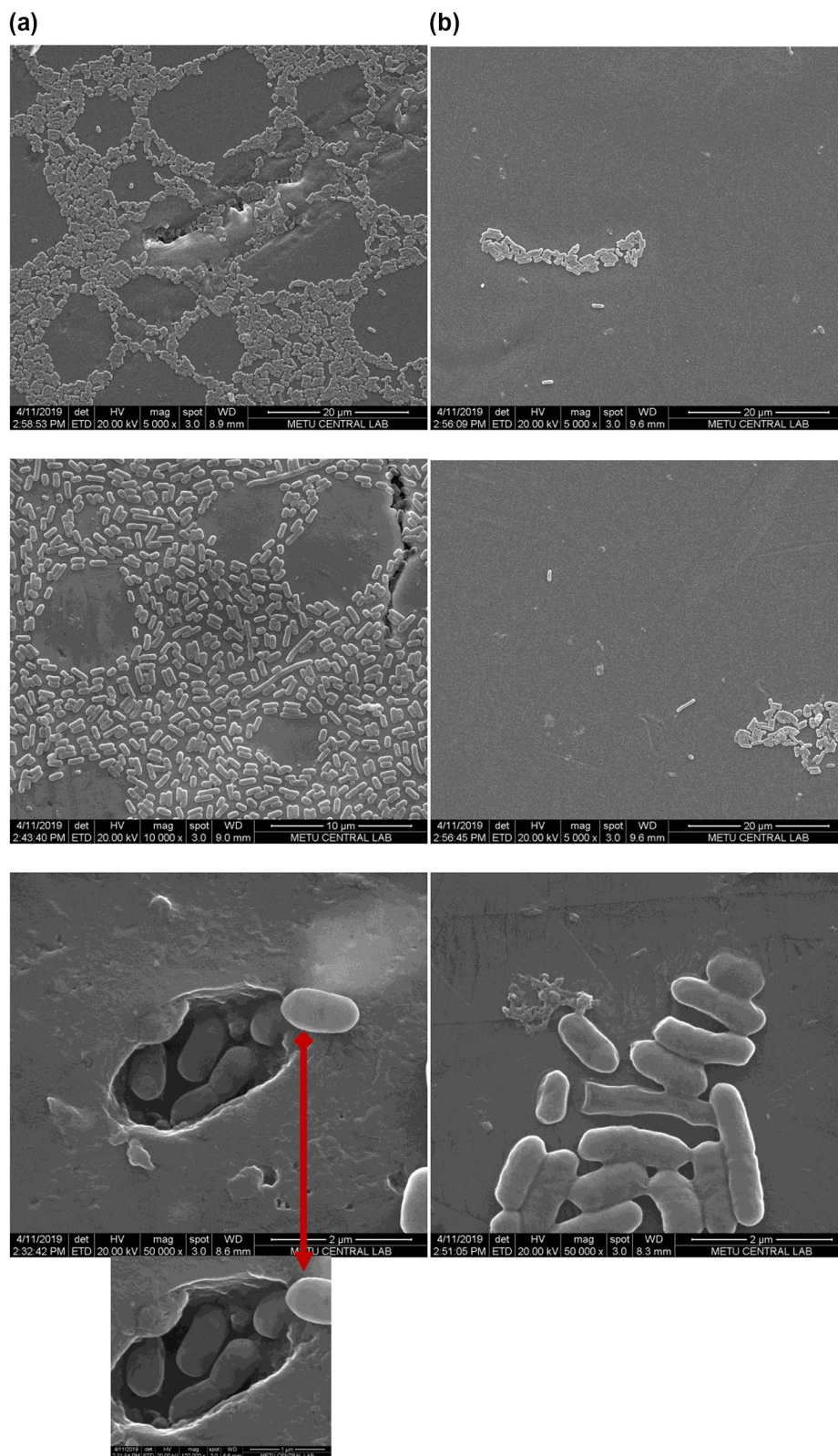


FIGURE 5 Scanning electron micrographs illustrating the biofilm formation of *L. monocytogenes* 32 on polycarbonate surface (column A: untreated; column B: treated with the sub-MIC value of CFE)

TABLE 3 Effect of *L. plantarum*-derived CFE on the cell viability of *L. monocytogenes* 32 in biofilm, formed on different surfaces (log CFU/cm²)

Surfaces	Untreated	Treated with sub-MIC CFE
316L Stainless steel surface (SS)	6.42 ± 0.15	1.99 ± 0.13 ^a
Polyvinyl chloride surface (PVC)	7.34 ± 0.05	2.12 ± 0.19 ^a
Polycarbonate surface (PC)	6.34 ± 0.45	2.49 ± 0.14 ^a

^aNote: Means with same letters in the same column are not significantly different ($p > .05$).

extracellular polymeric substance. A channel-like structure in the center of a “honeycomb” could also be seen in 50,000× magnification (Figures 3a–5a).

3.4 | Determination of the cell viability of *L. monocytogenes* 32 on different surfaces

The results obtained from a poultry isolate of *L. monocytogenes* 32 proved that this pathogen can be a contamination source with its strong biofilm formation ability. To achieve a new strategy to control this major problem in food processing environments, we used *L. plantarum*-derived CFE. *L. monocytogenes* 32 and CFE were co-incubated on different surfaces and the number of bacteria recovered from the biofilm was assessed. Following the co-incubation with CFE, reductions in the viable cell as log CFU/cm² were given in Table 3. The number of bacteria re-isolated from the biofilm on PVC surface was found as 7.34 ± 0.05 log CFU/cm², while 6.42 ± 0.15 log CFU/cm² and 6.34 ± 0.45 log CFU/cm² for PC and SS surfaces, respectively. The highest number of biofilm cells (5.22 log CFU/cm²) was achieved on PVC surfaces, but no statistical differences ($p > .05$) were found between the other surfaces (Table 3). Similar results were found with the combination of sonication and disinfection based on quaternary ammonium compounds (Wiktorczyk et al., 2019) and bacteriocin (Pérez-Ibarreche et al., 2017). Similar to our results, a disinfectant solution including plant essential oils obtained from *Cymbopogon citratus* and *Cymbopogon nardus* was reported to completely reduce the number of *L. monocytogenes* stainless steel surface-adhered cells (Oliveira et al., 2010). On the other hand, the phage P100 treatment significantly reduced *L. monocytogenes* cell populations under biofilm conditions on stainless steel coupon surface (Soni & Nannapaneni, 2010).

The SEM micrographs indicated not only a reduction in biofilm formation but also the decreasing of viable cell numbers on the treated samples compared to the control. Less attachment of bacterial cells to the surface was also observed on all surfaces in the treated samples compared to control (Figures 3b–5b). As a result of the CFE co-incubation, it is seen that the live cell count attached to the biofilms reduced. Based on this finding, live cells which have the potential of spreading can be countered by the use of CFE in the production of biofilms used in food processing.

4 | CONCLUSION

High costs, procurement difficulties, or resistance problems limit the practical use of several approaches focused on the inhibition of *L. monocytogenes*. Results obtained in this study will represent the first step in the prevention of biofilm formation using CFE obtained from *L. plantarum* which represents a safe and environmentally friendly approach to mitigate food-processing contamination.

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CONFLICTS OF INTEREST

The authors have declared no conflicts of interest for this article.

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